

D. Make Them Equal

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You have an array of integers a of size n . Initially, all elements of the array are equal to 1. You can perform the following operation: choose two integers i ($1 \leq i \leq n$) and x ($x > 0$), and then increase the value of a_i by $\lfloor \frac{a_i}{x} \rfloor$ (i.e. make $a_i = a_i + \lfloor \frac{a_i}{x} \rfloor$).

After performing all operations, you will receive c_i coins for all such i that $a_i = b_i$.

Your task is to determine the maximum number of coins that you can receive by performing no more than k operations.

Input

The first line contains a single integer t ($1 \leq t \leq 100$) — the number of test cases.

The first line of each test case contains two integers n and k ($1 \leq n \leq 10^3$; $0 \leq k \leq 10^6$) — the size of the array and the maximum number of operations, respectively.

The second line contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^3$).

The third line contains n integers $c_1, c_2, \dots, c_n$ ($1 \leq c_i \leq 10^6$).

The sum of n over all test cases does not exceed 10^3 .

Output

For each test case, print one integer — the maximum number of coins that you can get by performing no more than k operations.

Example

input	output
4	9
4 4	0
1 7 5 2	30
2 6 5 2	167
3 0	
3 5 2	
5 4 7	
5 9	
5 2 5 6 3	
5 9 1 9 7	
6 14	
11 4 6 2 8 16	
43 45 9 41 15 38	